THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 3: **World of Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The smallest natural number n for which $\frac{n}{490}$ has a terminating decimal expansion is: 1

(a) 50 **(b)** 51 **(c)** 49 **(d)** 48

2. If $x = 3 + \sqrt{8}$ with $3^2 - 8 = 1$, then $x + \frac{1}{x} =$ 1

(a) 6 **(b)** 5 **(c)** 7 **(d)** 8

3. $1.\overline{36}$ in the form $\frac{p}{q}$ is: 1

(a) $\frac{15}{11}$ **(b)** $\frac{5}{4}$ **(c)** $\frac{16}{11}$ **(d)** $\frac{14}{11}$

4. A number whose decimal expansion is $0.1010010001\ldots$ (the number of zeros increasing) is: 1

(a) a terminating decimal **(b)** irrational
(c) rational **(d)** an integer

5. The length of the repeating block of $\frac{1}{27}$ is: 1

(a) 2 **(b)** 3 **(c)** 4 **(d)** 5

6. $0.\overline{27} \times 11$ equals: 1

(a) 3 **(b)** $\frac{3}{2}$ **(c)** 4 **(d)** 2

7. The value of $0.\overline{3} + 0.\overline{7}$ is: 1

(a) $\frac{10}{9}$ **(b)** 1 **(c)** $\frac{10}{11}$ **(d)** $\frac{11}{9}$

8. $0.\overline{45} + 0.\overline{54}$ equals: 1

(a) 2 **(b)** 1 **(c)** 0 **(d)** $\frac{1}{2}$

9. **Assertion (A):** $(\sqrt{3} + \sqrt{2})(\sqrt{3} - \sqrt{2}) = 1$ is a rational number. 1
Reason (R): The product of two irrational numbers is always irrational.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The decimal expansion of $\frac{1}{7}$ is terminating. 1

Reason (R): $\frac{1}{7} = 0.\overline{142857}$, a non-terminating repeating decimal.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the smallest natural number n such that $\frac{n}{90}$ has a terminating decimal expansion. 2

12. If $x = 7 + \sqrt{48}$, find $\frac{1}{x}$ and $x + \frac{1}{x}$. 2

13. Express $0.\overline{36} + 0.\overline{63}$ as a single fraction in lowest terms. 2

14. Find the smallest natural number n such that $\frac{n}{1430}$ has a terminating decimal expansion. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Express $6.7\overline{9}$ as a fraction in lowest terms and verify by long division. 3

16. If $x = 4 + \sqrt{15}$, find $x^2 + \frac{1}{x^2}$. 3

17. Given that $\sqrt{5}$ is irrational, prove that $7 + \sqrt{5}$ is irrational. 3

OR

Express $0.9\overline{4}$ as a fraction in lowest terms and verify by long division. 3

18. If $x = 5 + \sqrt{24}$, find $x^2 + \frac{1}{x^2}$. 3

OR

Given that $\sqrt{6}$ is irrational, prove that $4 + \sqrt{6}$ is irrational. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Ritu is tiling her courtyard. One square tile has area 2 m^2 and another square tile has area 8 m^2. 5

(i) Write the side of the first tile.

(ii) Write the side of the second tile in the form $a\sqrt{2}$.

(iii) Find the product of the two sides.

(iv) Classify $5 + \sqrt{2}$ as rational or irrational with reason, and find $(\sqrt{2} + 1)(\sqrt{2} - 1)$.

20. Find the decimal expansion of $\frac{10}{37}$ by long division, state the repeating block and its length, and hence write $\frac{10}{37}$ as a repeating decimal. 5

OR

Simplify $(\sqrt{10} + \sqrt{6})^2 + (\sqrt{10} - \sqrt{6})^2$ and $(\sqrt{10} + \sqrt{6})(\sqrt{10} - \sqrt{6})$. Which are rational? 5

END OF QUESTION PAPER — **Total: 40 Marks**